Diagnostics

ACIC 2026 Short Course

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Link: https://tinyurl.com/acic2026-hte

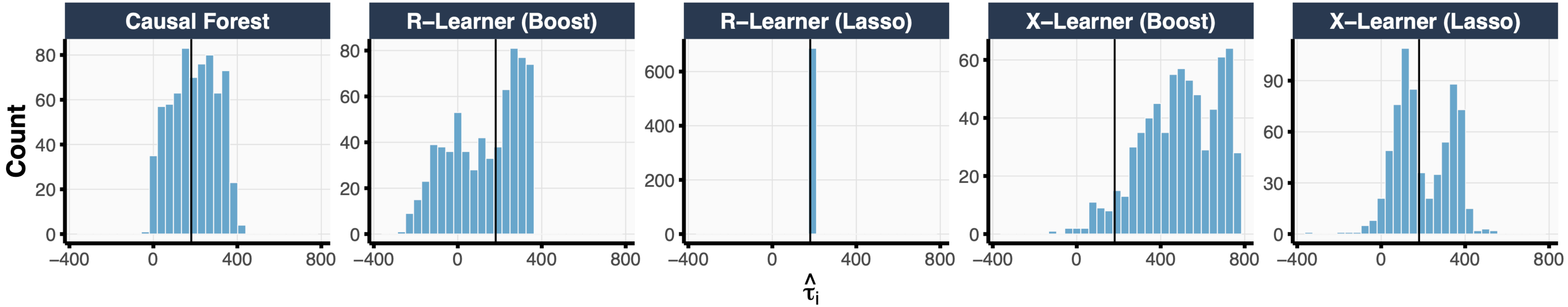
From Falco’s handoff

Falco closed with an end-to-end discovery workflow, and the output of that pipeline can vary depending on what downstream method is used after an IATE estimator.

Possible concerns:

- The IATE model itself is misspecified/a poor fit, discoveries are really just noise
- Another metalearner would have given a meaningfully different (possibly more accurate) answer

Motivation



- Using the same set of pre-treatment covariates, we fitted 5 metalearners on the GBSG2 breast cancer hormonal therapy trial to estimate the heterogeneous treatment effects
- The black line denotes the difference-in-means estimate of the overall ATE

> Which one do we report?

Two diagnostic questions today

1. Goodness-of-fit

How well are we approximating the underlying treatment effect heterogeneity?

→ Omnibus test

2. Model selection

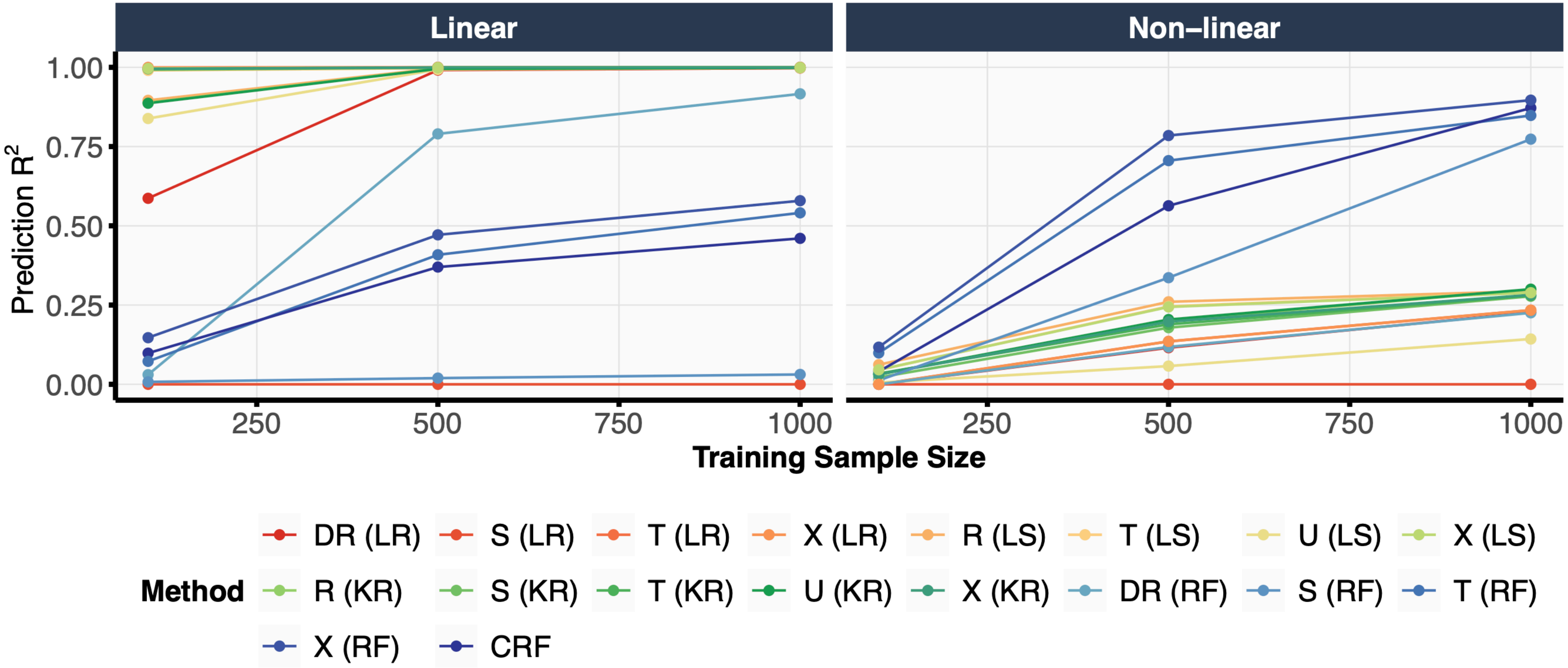
Among several candidate metalearners, which is best and how confident are we?

→ Relative error

The shared challenge: the individual treatment effect $\tau_i = Y_i(1) - Y_i(0)$ is **never observed**. Every diagnostic must be built around an *observable proxy*.

Omnibus test

The importance of model specification



Prediction R^2 vs training sample size across 18 metalearners under two DGPs. The best learner under a linear DGP is among the worst under a non-linear DGP — and vice versa. ***Picking a metalearner without knowing the DGP is structurally hard,*** *and we never know the DGP in practice.*

Why standard quality checks fail

Typically, in supervised learning, we can rely on a variety of metrics, e.g., test error.

For HTE estimation this **no longer applies**:

- Each unit reveals only $Y_i(T_i)$, never both $Y_i(1)$ and $Y_i(0)$
- $\tau_i = Y_i(1) - Y_i(0)$ is unobservable, so $(\hat{\tau}(X_i) - \tau_i)^2$ is uncomputable
- Comparing $\hat{Y}_i$ to Y_i doesn't catch the relevant kind of misspecification. **A metalearner can predict outcomes well and predict heterogeneity terribly.**

> **We need a test that uses observable quantities but targets the unobservable $\tau(X)$.**

The Best Linear Predictor (BLP) framing

Chernozhukov et al. (2018): project the true CATE onto $\hat{\tau}(X)$ via the BLP:

$$\mathrm{BLP}\left[\tau(X) \mid \hat{\tau}(X)\right] \;=\; \alpha_\tau + \beta_\tau\left\{\hat{\tau}(X) - \mathbb{E}[\hat{\tau}(X)]\right\}$$

where $\alpha_\tau = \mathbb{E}[\hat{\tau}(X)]$ and $\beta_\tau = \dfrac{\mathrm{cov}(\tau(X),\, \hat{\tau}(X))}{\mathrm{var}(\hat{\tau}(X))}$.

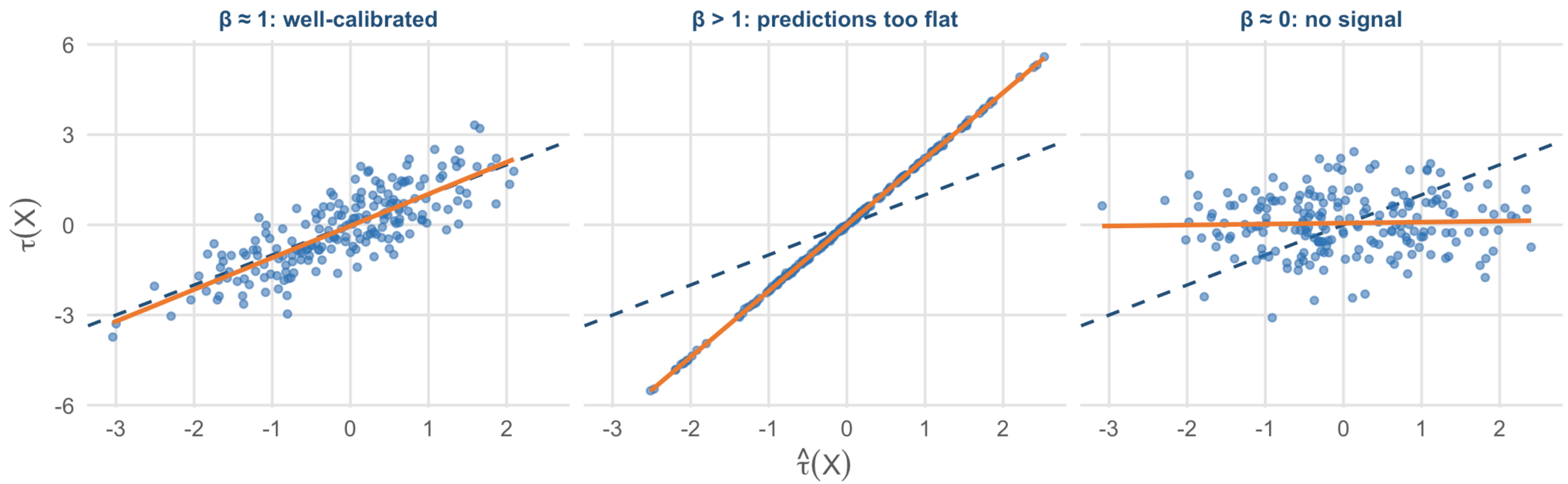
Re-write β_τ:

Let $v(X) = \tau(X) - \hat{\tau}(X)$. Then

$$\beta_\tau \;=\; 1 + \mathrm{cor}(v(X), \hat{\tau}(X)) \cdot \frac{\mathrm{sd}(v(X))}{\mathrm{sd}(\hat{\tau}(X))}$$

> $\beta_\tau \neq 1$ means residual error is *correlated* with the prediction. A sign of issues with the fit.

What different values of β_τ look like



Dashed line: $\beta_\tau = 1$ (perfect calibration). Orange: fitted slope of τ on $\hat{\tau}$.

Three regimes for β_τ.

Estimating β_τ in practice

Chernozhukov et al. (2018) show β_τ can be recovered from a Neyman-orthogonal regression:

$$Y_i = \hat{\gamma}_0 + \hat{\gamma}_x \hat{b}_0(X_i) \; + \; \hat{\alpha}_\tau\{Z_i - \hat{\pi}(X_i)\} \; + \; \hat{\beta}_\tau\{Z_i - \hat{\pi}(X_i)\}\big\{\hat{\tau}(X_i) - \hat{\mathbb{E}}[\hat{\tau}(X_i)]\big\} + \hat{\varepsilon}_i$$

where $\hat{b}_0$ is the control outcome model and $\hat{\pi}$ is the propensity score.

Test: $H_0 : \beta_\tau = 0 \;\; \text{vs} \;\; H_A : \beta_\tau > 0$

What "fail to reject" can mean:

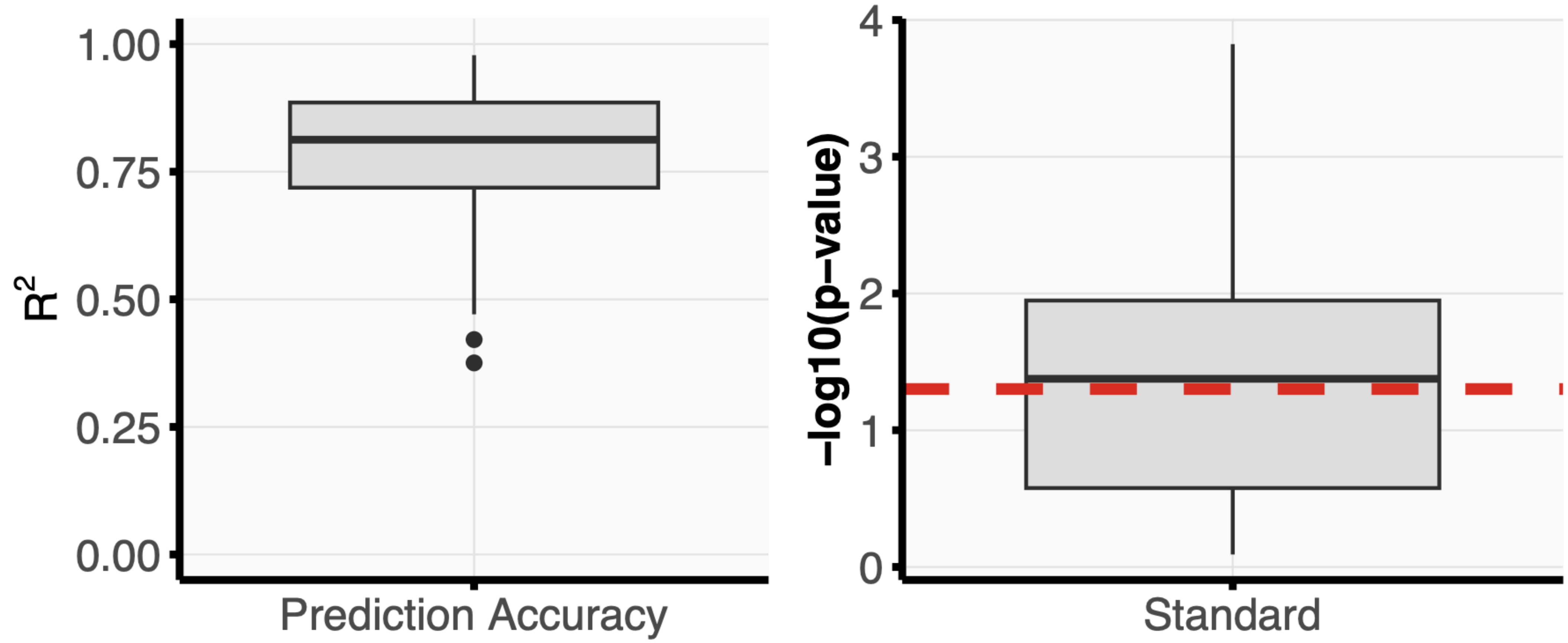
- $\tau(X)$ is genuinely constant (no heterogeneity to find), **or**
- $\hat{\tau}(X)$ is poorly estimated and the test simply has no power

Sample-splitting matters here too

The orthogonal regression on the previous slide uses *plug-in* estimates of $\hat{\pi}$, $\hat{b}_0$, and $\hat{\tau}$, they are all functions of the same data.

> Without sample splitting, asymptotic normality of $\hat{\beta}_\tau$ fails: residuals from the auxiliary fits leak into the test. Standard errors become unreliable.

Drawback: Sensitive to noise in outcomes



Example: Though $< 25\%$ of instances had R^2 values below 0.7, 47% of the tests were not significant.

What the test does — and doesn't — tell you

When the omnibus test rejects:

- You have evidence that $\hat{\tau}$ captures *some* signal
- The model is at least minimally calibrated to τ
- It does *not* tell you the model is right everywhere or that the discovered subgroups are correct

Next step: relative-error comparisons — is *this* the best model?

When the omnibus test fails to reject:

1. Inspect $\hat{\tau}$ histograms. Does the metalearner predict any heterogeneity at all?
2. If $\hat{\tau}$ is nearly constant: nothing for the test to detect.
3. If $\hat{\tau}$ varies but $\hat{\beta}_\tau$ is small: the heterogeneity in $\hat{\tau}$ is uncorrelated with truth: model-selection concerns

Relative error

The model-selection problem

In practice you fit multiple metalearners — say a causal forest, an X-learner with random forest, an R-learner with LASSO — and they disagree (recall the 5 stories from the breast-cancer figure).

> **Which one should you pick?**
>
> Without ground truth τ_i, you can't compute the obvious metric, test MSE on τ.

Goal: a procedure that

- works on observed data only (no oracle τ)
- yields a confidence set $\hat{S}$ for the *winning* estimator
- controls the probability that the true winner is missed at level α

Relative error

Instead of estimating $\mathbb{E}[(\hat{\tau} - \tau)^2]$ directly (which requires τ), estimate the *difference*:

$$\delta(\hat{\tau}_1, \hat{\tau}_2) \;=\; \mathbb{E}[(\hat{\tau}_1 - \tau)^2] - \mathbb{E}[(\hat{\tau}_2 - \tau)^2]$$

> The unobservable τ enters δ only through *one* term. We can estimate it.

Pairwise relative error (Gao, 2025)

For two estimators $\hat{\tau}_1, \hat{\tau}_2$, the difference in MSE is

$$\delta(\hat{\tau}_1, \hat{\tau}_2) \triangleq \mathbb{E}[(\hat{\tau}_1(X) - \tau(X))^2] - \mathbb{E}[(\hat{\tau}_2(X) - \tau(X))^2]$$

Expanding the squares:

$$\delta(\hat{\tau}_1, \hat{\tau}_2) = \mathbb{E}\left[\hat{\tau}_1^2(X) - \hat{\tau}_2^2(X)\right] - 2\,\mathbb{E}\left[(\hat{\tau}_1(X) - \hat{\tau}_2(X))\,\tau(X)\right]$$

Why this matters. The dependence on the unobserved τ is now **first-order**: $\tau(X)$ enters δ as an inner product against $(\hat{\tau}_1 - \hat{\tau}_2)$, not as a squared term.

→ Plug-in estimation of τ contributes only first-order error to $\hat{\delta}$.

→ First-moment relatively easier to estimate than second moment of τ

How to estimate

Use K-fold cross-fitting: train nuisances on $\mathcal{D}_{-k}$, evaluate $\hat{\delta}$ on $\mathcal{D}_k$, average.

Note: δ estimated following efficient influence functions in (Hines, Diaz-Ordaz, and Vansteelandt 2022)

Asymptotic normality (Gao, 2025). Under assumptions on overlap, consistency, and stability of the nuisance estimators,

$$\frac{\hat{\delta} - \delta}{\sqrt{\widehat{V}/n}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Extensions (Guo, Gao 2025): confidence sets

For K candidate estimators, we want a confidence set $\hat{S}$ for the winner $\hat{\tau}_{(1)}$.

For each candidate $r \in \{1, \dots, K\}$, test whether r dominates all competitors:

$$H_0^{(r)} : \delta(\hat{\tau}_r, \hat{\tau}_s) < 0 \;\; \forall s \neq r \quad \text{vs} \quad H_A^{(r)} : \exists s : \delta(\hat{\tau}_r, \hat{\tau}_s) > 0$$

Family-wise error rate (FWER) control. $\hat{S}$ contains the true winner with probability $\geq 1 - \alpha$:

$$\liminf_{n \to \infty} \mathbb{P}(\hat{\tau}_{(1)} \in \hat{S}) \geq 1 - \alpha.$$

Any candidate failing to reject $H_0^{(r)}$ is *kept* in $\hat{S}$.

Naive baseline: a max-statistic test

For each candidate r, build pairwise z-statistics and test against the *worst* one:

$$S_{r,s} = \frac{\hat{\delta}(\hat{\tau}_r, \hat{\tau}_s)}{\hat{\sigma}_{r,s}}, \qquad S_r^{\max} = \max_{s \neq r} S_{r,s}.$$

Add r to $\hat{S}$ if $S_r^{\max}$ is below its $1-\alpha$ null quantile.

The problem. The max-statistic treats every competitor symmetrically. As K grows:

A more powerful test: exponentially weighted aggregation

For estimator $\hat{\tau}_r$, build a *weighted* aggregate of pairwise relative errors per data point

> **Idea.** Up-weight competitors that look genuinely close to r; down-weight obviously-inferior competitors that would otherwise inflate the critical threshold.
>
> The exponential mechanism provides the right rate of weight decay.

Result: powerful argmin inference that adapts to which competitors are *close* to the winner, not how many candidates are in the set.

Wrap-up

RECAP & HANDOFF

Two diagnostics

Omnibus

Are we picking up a signal?

BLP-based $\hat{\beta}_\tau$ with proper cross-fitting/sample splitting; treat as a screen

Relative error

Compare candidate metalearners *without* ground truth, is this the best one?

Cross-fit one-step estimator and test

The shared message. Every diagnostic in this section is built around the fact that τ is unobservable, but observable proxies (BLP residuals, pairwise relative error) give us something to work off of.

Handoff to Tiffany: Hands-on Lab

What's next

Tiffany walks you through how to actually run all of this in R including: various metalearners, CDT, causal rule ensemble, feature importances, and some of the diagnostic methods we just discussed.

Thank you!

Additional slides

A more powerful test: exponentially weighted aggregation

For estimator $\hat{\tau}_r$, build a *weighted* aggregate of pairwise relative errors per data point:

$$Q_{i,r} = \sum_{j \neq r} \hat{\omega}_{r,j}\, \hat{t}(Z_i; \hat{\tau}_r, \hat{\tau}_j), \qquad \hat{\omega}_{r,j} \propto \exp\big(\eta\, \hat{\delta}(\hat{\tau}_r, \hat{\tau}_j)\big)$$

Idea. Up-weight competitors that look genuinely close to r; down-weight obviously-inferior competitors that would otherwise inflate the critical threshold.

The exponential mechanism provides the right rate of weight decay.

Result: powerful argmin inference that adapts to which competitors are *close* to the winner, not how many candidates are in the set.